

8th International Conference on Industry of the Future and Smart Manufacturing

Appendix A. An Ontology-Based Approach to Integrating Digital Product Passport Data into Enterprise Asset Management Systems

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Appendix A. Supplementary Research Methodology

This document contains the supplementary methodology data for the research paper: An Ontology-Based Approach to Integrating Digital Product Passport Data into Enterprise Asset Management Systems.

A.1. PICOC framework

Table A.1. PICOC framework.

Population (P)	Organizations utilizing Enterprise Asset Management (EAM) systems.
Intervention (I)	Integration of Digital Product Passport (DPP) data using semantic technologies and linked data principles.
Comparison (C)	Existing data management practices in asset management, manual file-based systems.
Outcome (O)	Improved asset traceability, data-driven decision-making, optimized life cycle management and asset management through DPP integration.
Context (C)	Organizations affected by the ESPR framework and implementing DPP data in EAM systems.

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A.2. Search Strategy

Table A.2. Search strategy, database, search field and search results/hits.

Search strategy	Database	Search field	Search Results / Hits
C1 AND C2	Scopus	Advanced Search	142
C1 AND C2 AND C3	Scopus	Advanced Search	125
C1 AND C2 AND C3 AND C4	Scopus	Advanced Search	77
C1 AND C2 AND C3 AND C6	Scopus	Advanced Search	45
C1 AND C2 AND C3 AND C4 AND C6	Scopus	Advanced Search	30
C1 AND C2 AND C3 AND C4 AND C5	Scopus	Advanced Search	13
C1 AND C2 AND C3 AND C4 AND C5 AND C6	Scopus	Advanced Search	5

A.3. Interview participants

Table A.3. Interview participants summary.

Interview participant	Experience (Y)	Role/Position	Company Type	Company Size*
A	9+ Years	Project Manager / Asset Manager	IT Consultancy	Small
B	35+ Years	Electrical Power and Lighting Maintenance Engineer	Government Agency	Large
C	29+ Years	Digital Aviation Director	Engineering, Advisory, and Consulting Services	Large
D	7+ Years	HVAC & Maintenance Utility Engineer	Engineering, Advisory, and Consulting Services	Large
E	4+ Years	Data Manager	Engineering, Advisory, and Consulting Services	Large

*In this study, company size is measured by the total number of employees. Small company = 10 to 49 employees. Large company = 1000 plus employees.

A.4. MOMo Workflow Methodology

Table A.4. MOMo workflow methodology steps, adapted from Shimizu et al. [1].

Step	Responsible	Output
1. Describe use cases & data sources	Entire team	Use case descriptions
2. Gather competency questions	Entire team	List of CQs
3. Identify key notions	Entire team	List of key notions
4. Identify existing ODPs	Ontology engineers	Selected Ontology Design Patterns, ODP(s) for each key notion.
5. Create module diagrams	Entire team	Diagrammatic representation of the solution module.
6. Document modules & axioms	Ontology engineers & domain experts	Module documentation with embedded schema diagrams, axiomatization, etc. (e.g., in $\text{\LaTeX}$, Word, HTML format).
7. Create ontology diagram	Ontology engineers	Diagrammatic representation of the whole composed ontology.
8. Add spanning axioms	Ontology engineers	Documentation of the entire ontology with embedded schema diagrams, axiomatization, etc. (e.g., in $\text{\LaTeX}$, Word, HTML format).
9. Review naming & axioms	Ontology engineers	Updated module and ontology documentation.
10. Create OWL file & axioms	Ontology engineers	An OWL file for publication and use.

A.5. PRISMA flowchart

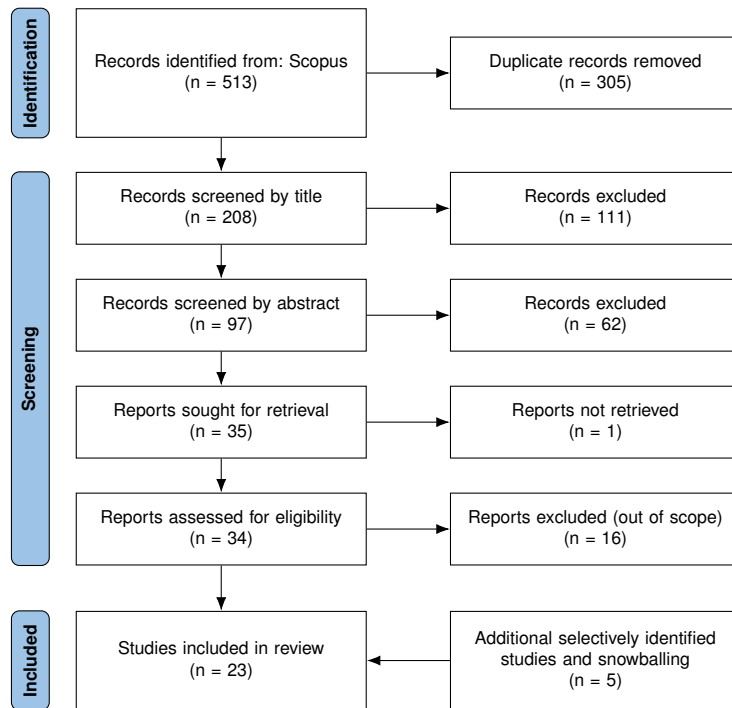


Fig. A.1. PRISMA flowchart.

References

- [1] Shimizu, C., K. Hammar, and P. Hitzler. (2023) “Modular ontology modeling.” *Semantic Web* **14** (3): 459–489.